

Integral periodic orbits of a cubic three-term recurrence

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Abstract

We classify every integral periodic orbit of $T_a(x, y, z) = (y, z, yz + a - x)$ for every integer a . The answer consists of eight explicitly parametrized families and two sporadic orbits, with least periods exactly 1, 2, 3, 4, 5, 6, 8, 9, 12. The proof is computer-assisted, but its reduction is uniform in the parameter, invariant level and period. For a periodic scalar sequence, the two-step differences satisfy a linear identity in which the forcing parameter cancels. Extremal differences isolate the unbounded families; every remaining orbit has maximum difference at most 100 and all coordinates between -301 and 299 . A terminating exact certificate exhausts that residual set without a period cutoff. An independently organized, inverse-time, two-sign enumeration agrees on all 25 851 oriented cycles in the finite certificate. We resolve all least-period degeneracies and cyclic identifications, and give finite square tests for the number of oriented cycles on each integral invariant level. These yield the exact ordinary fixed-point counts and a finite rational dynamical zeta product. The classical unforced families and the earlier unforced integral classification are explicitly excluded from the contribution claimed here.

1 Introduction

Consider the polynomial automorphism

$$T_a(x, y, z) = (y, z, yz + a - x), \quad a \in \mathbb{Z}. \quad (1)$$

We ask for every periodic point in $\mathbb{Z}^3$, with one application of T_a as one time step. The forcing parameter is arbitrary and is held fixed along the orbit. A classification of short returns in chosen boxes would not answer this question: both the parameter and the height of periodic points are unbounded. In fact, there are infinitely many four-cycles for each fixed parameter when the invariant level is allowed to vary.

The main result gives a disjoint list of all oriented cycles. Its proof has two parts. First, a two-step difference identity removes a from the local growth constraints. A large coordinate then forces a four-cycle, while a large extremal difference leaves only five possible central coordinates. Their exact analysis extracts the other unbounded families. Second, the remaining difference and height bounds define a finite seed problem. Exact iteration from each seed stops at an exit or a first return by finiteness and injectivity; no maximum period is imposed. The two sporadic cycles are the remainder after the complete finite output is matched with the symbolic families. The finite computation is a proof component, not empirical support for a conjectured bound.

Prior work and the precise scope of the increment. The ambient cubic surfaces and their Vieta-type automorphisms are classical. Under $(X, Y, Z) = (-x, -y, -z)$ the invariant equation becomes

$$X^2 + Y^2 + Z^2 + XYZ = -a(X + Y + Z) + k,$$

a specialization of the standard character-variety cubic family studied by Cantat and Loray [2]. For $a = 0$, the map is the Fibonacci trace map in full-trace coordinates. Its invariant, reversibility, and explicit periodic families are part of the trace-map theory of Roberts and Baake [6]; general escape mechanisms for unforced trace maps are developed by Roberts [5]. We do not claim these structures, the known four-step family, or the unforced periodic families as new. The anonymous predecessor [1] already gives the complete integral classification at $a = 0$; that entire subcase is likewise deducted here. None of its proof is needed as an unprinted lemma in the argument below.

Other trace-map finiteness questions have different quantifiers. The finite-orbit results for character-variety group actions in Cantat and Loray [2], and the comparison between finite group orbits and periodicity under every group element in Humphries [4], do not directly classify integer periodic points of this one specified map. The versioned Cantat–Loray preprint used for the group-action comparison is [arXiv:0711.1579v2](https://arxiv.org/abs/0711.1579v2); its Theorem C is not cited as a theorem number of the shorter journal layout. Hone’s related third-order recurrence [3, Proposition 1] has a parameter multiplying the product term, rather than the additive forcing in (1). These related settings motivate the methods but do not replace the all-forcing exhaustion proved here.

The contribution is the uniform extremal-difference reduction, its complete residual certificate, and the resulting all-parameter oriented classification and level counts. We make no worldwide priority claim. Table 1 records the scope distinctions that matter to the theorem.

Table 1: Source scope and the additional question answered here. The distinctions concern the object and quantifiers, not a priority ranking. All present counts use ordinary integer points and the specified map’s time step.

Input or related setting	Scope retained or distinguished
Classical cubic and trace maps	Ambient cubic, reversibility, unforced families, four-step family and escape framework are prior structures.
Unforced integral predecessor	The complete $a = 0$ classification is already established; it is reproved where needed here.
Whole-group finite orbits	A group orbit, or periodicity under every group element, is not periodicity under one fixed T_a .
Multiplicative-parameter recurrence	Its product coefficient is not the additive integer forcing a .
Present theorem	Every integer a , every height and period; all oriented cycles and exact invariant-level multiplicities.

Organization. Section 2 states the full list. Sections 3 and 4 prove the uniform analytic reduction. Section 5 specifies and justifies both exact finite certificates. Section 6 verifies least periods and representatives, and Section 7 derives the arithmetic level tests. The complete primary certifier and evidence identifiers are included in Appendix A.

2 The complete classification

The inverse and a polynomial invariant of (1) are

$$T_a^{-1}(x, y, z) = (xy + a - z, x, y), \quad (2)$$

$$K_a(x, y, z) = x^2 + y^2 + z^2 - xyz - a(x + y + z). \quad (3)$$

The inverse identity is immediate. For invariance, put $w = yz + a - x$. The difference $K_a(y, z, w) - K_a(x, y, z)$ factors as $(w - x)(w + x - yz - a) = 0$.

A repeated word $w = (w_0, \dots, w_{\ell-1})$ encodes the consecutive triples (w_i, w_{i+1}, w_{i+2}) . These form an orbit, with indices modulo ℓ , provided

$$w_{i+3} = w_{i+1}w_{i+2} + a - w_i \quad \text{for every } i \pmod{\ell}. \quad (4)$$

Conversely every periodic triple orbit has this encoding. Its least scalar period equals the first-return time of a triple: a repeated triple determines the entire forward and backward sequence by (1) and (2). An *oriented cycle* is an orbit under forward iteration. Cyclic rotations represent the same cycle. We do not identify a word with its reverse unless those two words are cyclic rotations.

Theorem 2.1 (All integral periodic orbits). *For every $a \in \mathbb{Z}$, all periodic points of T_a in $\mathbb{Z}^3$ are the consecutive triples of precisely the words in Table 2. The indicated parameter ranges give one representative per oriented cycle, with exactly the stated least period and invariant level. The table includes singular points of every level surface.*

Table 2: Complete disjoint list of oriented integral cycles. Every displayed word is repeated. The range conditions remove cyclic duplications and shorter-period cases; reversal is not an additional quotient. The symbol k denotes K_a .

ℓ	Word and parameter range	k
1	$F_1 : (r); r \in \mathbb{Z}, a = 2r - r^2$	$r^2(2r - 3)$
2	$F_2 : (u, v); u < v, a = u + v - uv$	$uv(u + v - 3)$
3	$F_3 : (-2, -2, t); a = 2t - 4, t \neq -2$	$8 - (t - 4)^2$
4	$F_4 : (-1, t, -1, a + 1 - t); 2t < a + 1$	$t^2 - (a + 1)t + 2a + 2$
5	$F_5 : (-1, t, 0, 0, -1 - t); a = -1, t \in \mathbb{Z} \setminus \{-1\}$	$t(t + 1)$
5	$E_5 : (-4, -2, -3, -3, -2); a = -13$	-64
6	$F_6 : (0, 0, t, 0, 0, -t); a = 0, t \geq 1$	t^2
8	$F_8 : (-1, t, 1, t, -1, -t - 2, 1, -t - 2); a = -1, t \geq 0$	$(t + 1)^2 + 1$
9	$E_9 : (-2, -1, 0, 0, -1, -2, 0, -1, 0); a = -2$	-1
12	$F_{12} : (1, m, 1, m - 1, -1, -m, 1, 1 - m, 1, -m, -1, m - 1); a = 0, m \geq 1$	$m^2 - m + 2$

There are eight parametric rows and two sporadic rows. The labels F_ℓ denote the parametric families and E_5, E_9 the two exceptions. In particular, the set of possible least periods is

$$\mathcal{L} = \{1, 2, 3, 4, 5, 6, 8, 9, 12\}. \quad (5)$$

Each length actually occurs: one may take $r = 0$ in F_1 , $(u, v) = (0, 1)$ in F_2 , $t = 0$ in F_3 , $(a, t) = (0, 0)$ in F_4 , $t = 0$ in F_5 , $t = 1$ in F_6 , $t = 0$ in F_8 , the word E_9 , and $m = 1$ in F_{12} . Thus T_a^{360} fixes the integral periodic locus for each a . This is not a finite-order assertion about T_a on its full phase space.

The theorem's exhaustion is computer-assisted: the analytic part reduces every unlisted cycle to a proved finite core, whose complete exact evaluation is described in Section 5. All existence, period and overlap statements for the infinite families are polynomial or finite-word arguments and apply outside that core as well.

3 A parameter-free difference identity

Let $(x_i)_{i \in \mathbb{Z}}$ be a periodic integral sequence satisfying (4). Define

$$d_i = x_{i+2} - x_i, \quad D = \max_i |d_i|. \quad (6)$$

Subtract the recurrence at $i - 1$ from that at i . It gives

$$d_{i-1} + d_{i+1} = (x_{i+1} + 1)d_i, \quad |(x_{i+1} + 1)d_i| \leq 2D. \quad (7)$$

The forcing cancels. Two adjacent zero differences force all differences to be zero, by propagation of (7) in both directions.

Reversal of a scalar sequence preserves (4). On triples it is the actual reversing involution

$$R(x, y, z) = (z, y, x), \quad RT_a R = T_a^{-1}. \quad (8)$$

We use this symmetry to normalize an extremum, but restore both time orientations at the end.

Lemma 3.1 (Zero difference). *If $D = 0$, the scalar sequence alternates two integers u, v and*

$$a = u + v - uv = 1 - (u - 1)(v - 1). \quad (9)$$

It is fixed when $u = v$ and otherwise has least period two.

Proof. The equation $d_i = 0$ is $x_{i+2} = x_i$. Substitution of $(u, v, u, v, \dots)$ into the recurrence is equivalent to (9). The assertion about the least period follows directly. $\square$

Lemma 3.2 (The high-coordinate four-step channel). *If $D > 0$ and $|x_j + 1| > 3D$ at some phase, the orbit is a word of the form $(-1, t, -1, a + 1 - t)$ repeated. Consequently, every orbit not of this form satisfies*

$$|x_i + 1| \leq 3D \quad \text{for all } i. \quad (10)$$

Proof. Rotate to $j = 1$. Integrality in (7) forces $d_0 = 0$. Put $x_0 = x_2 = b$, $x_1 = m$, and $e = d_1 = -d_{-1}$. We have $e \neq 0$, since two adjacent zero differences would give $D = 0$, and $|e| \leq D$. Moreover, $x_{-1} = x_3 = m + e$; in particular the sign in both of these expressions is the same. Thus $|x_{-1} + 1|, |x_3 + 1| > 2D$, forcing $d_{-2} = d_2 = 0$ by (7). Its equation at $i = 1$ is now $(b + 1)e = 0$, so $b = -1$. Direct use of the scalar recurrence yields $(-1, m, -1, a + 1 - m)$ repeated. Conversely this word satisfies every recurrence equation. Its period is four unless $2m = a + 1$, when it is alternating (and fixed if also $m = -1$). $\square$

4 Exhaustion when the maximum difference is large

Proposition 4.1 (Uniform reduction). *Every integral periodic sequence with $D > 100$ is one of the families $F_3, F_4, F_5, F_6, F_8, F_{12}$, allowing the shorter-period degeneracies before choosing the ranges in Table 2.*

Choose a phase and initially reverse if necessary so that $d_0 = D > 0$. Write

$$(x_0, x_1, x_2) = (u, s, v), \quad v - u = D, \quad p = d_{-1} = v - su + s - a, \quad q = d_1 = sv + a - u - s. \quad (11)$$

Equation (7) gives

$$s \in \{-3, -2, -1, 0, 1\}, \quad p + q = (s + 1)D, \quad |p|, |q| \leq D. \quad (12)$$

All bounds below are consequences of the global maximum D ; a violation excludes a periodic sequence, not just a sampled initial condition.

4.1 The endpoint centers

If $s = 1$, then $p = q = D$, and (7) gives $d_{-2} = uD$, $d_2 = vD$. Hence $u, v \in [-1, 1]$, implying $D \leq 2$, a contradiction. If $s = -3$, then $p = q = -D$ and $d_{-2} = -(u+2)D$, $d_2 = -(v+2)D$. Now $u, v \in [-3, -1]$, giving the same contradiction.

4.2 The center $s = -1$

Set $E = u + v - a - 1$, so $p = E$ and $q = -E$. We have

$$d_{-2} = (u+1)E - D, \quad d_2 = -(v+1)E - D.$$

Their bounds imply $(u+1)E \geq 0$ and $(v+1)E \leq 0$. Since $u < v$, the case $E > 0$ is impossible. If $E = 0$, the recurrence is the four-step word F_4 (up to rotation). Otherwise write $E = -\kappa$, with $\kappa \in \mathbb{Z}_{>0}$. Then

$$u \leq -1 \leq v, \quad \kappa(-u-1) \leq 2D, \quad \kappa(v+1) \leq 2D.$$

Adding the latter inequalities gives $\kappa \leq 4$. Both x_{-1} and x_3 equal $\kappa - 1$, and one more application of (7) gives

$$d_{-3} + d_3 = \kappa(\kappa - 2)D.$$

Its absolute value is at most $2D$, excluding $\kappa = 3, 4$.

For $\kappa = 1$ we have $a = u + v$, $x_{-1} = x_3 = 0$, $d_{-3} = -v$ and $d_3 = u$. The centers of these differences are both $a + 1$, so

$$|a + 2| \max(|u|, |v|) \leq 2D.$$

Since the maximum is at least $D/2$, it follows that $-6 \leq a \leq 2$. The forward sequence begins

$$u, -1, v, 0, a + 1, u, (a + 1)u + a, (a + 1)u^2 + au - 1.$$

Consequently $d_5 = (a + 1)u^2 + (a - 1)u - 1$. If $a \neq -1$, put $U = |u|$. As $u = (a - D)/2 < 0$, we have $U > 49$ and $D \leq 2U + 6$, while

$$|d_5| \geq U^2 - 7U - 1 > 2U + 6 \geq D,$$

a contradiction. If $a = -1$, the repeated word is $(u, -1, -1 - u, 0, 0)$, a rotation of F_5 .

For $\kappa = 2$ we have $a = u + v + 1$, $d_{-3} = -2a$ and $d_3 = 2a$, with centers $u + a + 1$ and $v + a + 1$. Applying (7) at these centers gives

$$|2a(u + a + 2)| \leq 2D, \quad |2a(v + a + 2)| \leq 2D.$$

The shifted coefficients differ by D , so one has absolute value at least $D/2$. Hence $|a| \leq 2$, including $a = 0$. The forward sequence begins

$$u, -1, v, 1, v + a + 1, 2a + 1, (v + a + 1)(2a + 1) + a - 1,$$

and hence

$$d_4 = 2a(v + a + 1) + a - 1, \quad D = 2v - a + 1, \quad v > 48.$$

For $a = 1$ we obtain $d_4 = D + 4$. For $a = 2$ we obtain $d_4 = 4v + 13 > D$, and for $a = -2$ we obtain $|d_4| = 4v - 1 > D = 2v + 3$. All are impossible. For $a = -1$ direct iteration gives F_8 ; for $a = 0$ it gives F_{12} . In the latter case $u = -1 - v$ and $m = v + 1 > 0$; the triple $(u, -1, v)$ occurs starting at position 9 of the displayed F_{12} word, with zero-based indices.

4.3 The signed normalization at $s = 0, -2$

For either remaining center, $|p + q| = D$, so at least one of $|p|, |q|$ is at least $D/2$. Reverse if needed to arrange $|p| \geq D/2$. This second reversal can change the extremum's sign. We therefore write

$$d_0 = \varepsilon L, \quad L = D > 100, \quad \varepsilon \in \{-1, 1\}, \quad v = u + \varepsilon L. \quad (13)$$

The bound at p gives $|u + 1| \leq 4$, so $-5 \leq u \leq 3$. Since $|v + 1| \geq L - 6$, the bound at q gives $|q| \leq 2L/(L - 6) < 3$. Keeping ε in the next two cases is essential.

4.4 The center $s = 0$

Here $p + q = \varepsilon L$, so p, q have sign ε or are zero. Write $q = \varepsilon r$, $r \in \{0, 1, 2\}$. Then $a = u + \varepsilon r$, $x_3 = \varepsilon r$, and

$$d_2 = (v + 1)\varepsilon r - \varepsilon L.$$

If $\varepsilon = -1$ and $r \geq 1$, this is $(r + 1)L - r(u + 1) > L$, a contradiction. If $\varepsilon = 1, r = 2$, it is $L + 2u + 2 \geq L - 8$, with center $x_3 = 2$. Equation (7) would require $3(L - 8) \leq 2L$, again impossible.

If $\varepsilon = 1, r = 1$, put $a = u + 1$, so $-4 \leq a \leq 4$. We obtain $d_2 = a$, $d_3 = 2a - 1$, with the latter center $x_4 = L + 2a - 1$. Thus

$$|(L + 2a)(2a - 1)| \leq 2L.$$

For an integer $a \notin \{0, 1\}$ the left side is at least $3(L - 8) > 2L$. If $a = 1$, the initial triple is $(0, 0, L)$, and $x_4 = L + 1$, $x_6 = 2L + 2$, so $d_4 = L + 1 > D$. If $a = 0$, the initial triple is $(-1, 0, L - 1)$, and $x_7 = 1 - L$, $x_9 = 1 - (L - 1)^2$, giving $d_7 = -L^2 + 3L - 1$ of magnitude greater than L .

Only $r = 0$ remains. Now $a = u$, $x_3 = 0$, $x_4 = u$, $x_5 = -\varepsilon L$, and $d_2 = d_3 = -\varepsilon L$. The bound at d_3 gives $-3 \leq u \leq 1$. Also $d_4 = -u\varepsilon L$, centered at $x_5 = -\varepsilon L$. Since $L > 100$, the inequality $|(-\varepsilon L + 1)u\varepsilon L| \leq 2L$ forces $u = 0$. Hence $a = 0$ and the word is F_6 .

4.5 The center $s = -2$

Here $p + q = -\varepsilon L$, so write $q = -\varepsilon r$, $r \in \{0, 1, 2\}$. Then

$$d_2 = -rL - \varepsilon(r(u + 1) + L).$$

For $\varepsilon = 1, r \geq 1$ this is at most $-(r + 1)L + 4r < -L$. Consider $\varepsilon = -1$. Then $q = r$, $v = u - L$, $a = 2v + u + r - 2$, $x_3 = r - 2$, and $x_4 = rv + u + r$.

If $r = 2$, we have $x_4 = 2v + u + 2$, $x_5 = v + u$, $a = 2v + u$, and $d_4 = x_4x_5 + a - x_4$. Using $|u| \leq 5$ gives the conservative estimate

$$|d_4| \geq (2L - 17)(L - 10) - 4L - 32 > L \quad (L > 100),$$

a contradiction. If $r = 1$, the next entries are

$$\begin{array}{lll} x_3 = -1, & x_4 = v + u + 1, & x_5 = -2, \\ x_6 = -u - 2, & x_7 = v + 2u + 2, & x_8 = -uv - 2u^2 - 5u - 3. \end{array}$$

Thus $d_5 = -L + 3u + 4$, centered at $-u - 2$. For $u \in [-5, 3] \setminus [-3, 1]$, $|(u + 1)d_5| \geq 3(L - 13) > 2L$, impossible. For $-3 \leq u \leq 1$, the difference $d_6 = uL - 3u^2 - 4u - 1$ has center $x_7 = -L + 3u + 2$. Its bound implies

$$|d_6| \leq \frac{2L}{L - 6} < 3.$$

If $u \neq 0$, then $|d_6| \geq L - 16 > 2$, contradicting integrality and this strict bound. For $u = 0$, direct iteration instead gives $d_7 = 2L - 7 > L$.

Only $r = 0$ remains, for either sign. Now

$$\begin{aligned} a &= 2v + u - 2, & x_3 &= -2, & x_4 &= u, & x_5 &= \varepsilon L - 2, \\ d_2 &= -\varepsilon L, & d_3 &= \varepsilon L, & d_4 &= (u + 2)\varepsilon L. \end{aligned}$$

The bound for d_4 at center x_5 forces $u = -2$. The resulting word is $(-2, -2, t)$ with $a = 2t - 4$, namely F_3 .

4.6 Restoring orientation and concluding the reduction

At the centered triple, reversal sends (d_0, p, q) to $(-d_0, -q, -p)$, exactly the rule used in (13). Reversing F_3, F_4, F_6, F_8 or F_{12} gives a rotation of the same word. For the form $(u, -1, -1 - u, 0, 0)$ of F_5 , the reverse is a rotation of $(-1 - u, -1, u, 0, 0)$, which is also included as the parameter ranges over all integers. Alternating words are reversal-closed as well. Thus normalization has not lost an oriented orbit. This completes the proof of Proposition 4.1.

Corollary 4.2 (The residual core). *Every integral periodic orbit is either an alternating word, one of $F_3, F_4, F_5, F_6, F_8, F_{12}$, or satisfies*

$$1 \leq D \leq 100, \quad -3D - 1 \leq x_i \leq 3D - 1 \quad \text{at every phase.} \quad (14)$$

In particular the residual coordinates belong to $[-301, 299]$.

Proof. Apply Lemma 3.1, Proposition 4.1, and then Lemma 3.2 outside F_4 . □

5 A terminating finite certificate

For an orbit remaining after the listed channels have been removed, the bound $D \leq 100$ is now a theorem, not a search parameter. This section specifies the complete finite task and records its accepted exact evaluation. The two enumerations were performed in the proof stage; they were not rerun merely to prepare this article. Appendix A prints the primary implementation and identifies the preserved source and output files.

5.1 The forward, positive-extremum algorithm

For each integer $D \in [1, 100]$ put

$$b_- = -3D - 1, \quad b_+ = 3D - 1.$$

Perform the following finite loops and exact orbit walks.

1. For every integer $u \in [b_-, b_+ - D]$ and every $s \in \{-3, -2, -1, 0, 1\} \cap [b_-, b_+]$, put $v = u + D$. For every integer $p \in [-D, D]$, put $q = (s + 1)D - p$. Retain the seed precisely when

$$|q| \leq D, \quad |(u + 1)p| \leq 2D, \quad |(v + 1)q| \leq 2D. \quad (15)$$

Set $a = v - su + s - p$ and the starting triple $P = (u, s, v)$. The code intersects the corresponding integer p intervals before entering their innermost loop; this is exactly the same finite set.

2. Keep this a, D fixed. At the current state (x, y, z) , test first whether a coordinate is outside $[b_-, b_+]$, then whether $|z - x| > D$. If either test holds, record the corresponding exit and stop this walk.
3. Otherwise append x to the scalar word and replace the state by $(y, z, yz + a - x)$. If it equals P , record the first return and stop; otherwise repeat the preceding step.
4. For each returned word, check the recurrence at every wraparound phase and absence of any smaller scalar period. Insert both the word and its reverse, identifying words only by cyclic rotation. Store the pair consisting of a and the lexicographically least rotation.
5. Compare each resulting word, at every cyclic phase, with the explicit family formulas in Table 2. For this subtraction allow their degenerate parameter values; classify by the actual word length. Output every unmatched word with its parameter and first-return length.

Lemma 5.1 (Coverage and termination). *The algorithm covers every orbit satisfying (14), including its original time orientation. Every orbit walk stops, without any imposed bound on a or on the number of iterates. An exit excludes the seed from a periodic orbit having the specified maximum difference and height bounds.*

Proof. Every residual periodic word has a phase with $|d_0| = D$; reverse if necessary to obtain $d_0 = D$. Equation (7) gives precisely the five choices of s , the relation defining q , and the two neighboring bounds (15). The height bounds place u, v in the enumerated intervals. The equation defining p in (11) uniquely recovers a . Thus an extremal seed of this word occurs. The reversed word is also a valid word for the same a by (8); explicitly restoring both reverses recovers whichever orientation was used originally.

For fixed a, D , all nonexiting states belong to the finite set $[b_-, b_+]^3 \cap \mathbb{Z}^3$. If a walk never exited, it would repeat a state. Under an injective map the first repeat must be the initial state: if $P_j = P_i$ with $j > i > 0$, apply the inverse i times to obtain $P_{j-i} = P_0$, an earlier return. This proves termination at a first return or exit. A genuine bounded periodic word could violate neither necessary bound, so an exit is a valid exclusion. The existence of this finite-state argument is not an operational period cutoff in the code. $\square$

5.2 Independent interval seeds and inverse traversal

The independent implementation changes the seed order, treats both extremum signs directly, walks the inverse map, and compares complete triple-state sets. It neither imports nor executes the primary implementation. Its complete mathematical algorithm is as follows.

For $D = 1, \dots, 100$, $\delta \in \{-D, D\}$, $s = -3, -2, -1, 0, 1$, and $q = -D, \dots, D$, set $p = (s + 1)\delta - q$ and discard $|p| > D$. Start with the integer interval

$$I = [b_-, b_+] \cap [b_- - \delta, b_+ - \delta]. \quad (16)$$

If $p \neq 0$, intersect I with

$$[-1 - \lfloor 2D/|p| \rfloor, -1 + \lfloor 2D/|p| \rfloor]; \quad (17)$$

if $q \neq 0$, intersect it with

$$[-1 - \delta - \lfloor 2D/|q| \rfloor, -1 - \delta + \lfloor 2D/|q| \rfloor]. \quad (18)$$

Zero p or q imposes no corresponding interval constraint. For each integer $u \in I$, put

$$v = u + \delta, \quad a = q - sv + u + s, \quad P = (u, s, v). \quad (19)$$

Check the reconstruction of both p and q and their bounds. All five centers already lie in $[b_-, b_+]$ for $D \geq 1$. These interval intersections are exactly the integer solutions of the same neighboring inequalities, now with both signs.

At each inverse-time state, test height first, then the difference bound, and then membership in a visited set. A repeated state must equal P ; a noninitial first repeat raises an explicit error. Otherwise insert the state and apply $(x, y, z) \mapsto (xy + a - z, x, y)$, checking that a forward step recovers the previous state. The proof of Lemma 5.1, applied to the inverse bijection, proves termination here too.

Upon return, choose the lexicographically least visited triple and reconstruct its native *forward* word. Check every forward state belongs to the inverse visited set, that the first repeat is the chosen triple, that all visited states are used, and that the word has as many distinct triples as entries. Represent this cycle by $(a, \min\{\text{visited triples}\})$, while retaining the complete triple set. For fixed a that set determines the forward transitions uniquely, so this representation does not quotient by reversal.

For independent family matching, generate candidate words rather than use the primary rotation predicates. Generate F_3 with $t = (a + 4)/2$ when a is even. For every value t occurring in the reconstructed word, generate F_4 and, at their required parameters, F_5, F_6, F_8 . At $a = 0$ also generate F_{12} with $m = \max_i |x_i| \geq 1$. For a word of length at most two, generate its alternating candidate. Verify candidate recurrences and compare their complete triple-state sets. Assign the first matching label in the order $F_2, F_3, F_4, F_5, F_6, F_8, F_{12}$; this resolves the degenerate family overlaps. An unmatched word is an exception. These candidate rules cover every family match: its parameter is either fixed by a , appears in the word, or equals its maximum absolute coordinate in F_{12} .

Only after discovery is complete does the independent algorithm parse the primary cycle output. It validates every recorded integer word, recurrence, first-return length and family label, rejects duplicate keys, compares the complete key sets in both directions, and compares every full triple-state set. Reading the input bytes earlier to hash them does not influence discovery. Both implementations use exact unbounded integers.

5.3 The complete finite outcome

The preserved executions give Table 3. Both implementations produce exactly 25 851 oriented cycles, with the identical family partition

$$(F_3, F_4, F_5, F_6, F_8, F_{12}, \text{unmatched}) = (200, 25350, 99, 100, 50, 50, 2). \quad (20)$$

These counts refer only to the finite certificate, not to the infinite global families. There are 25 753 self-reversing cycles and 49 pairs of distinct reversed cycles in this output; the latter all lie in F_5 .

Table 3: Complete historical seed dispositions in the proved residual core. The extremum sign and traversal direction are distinct choices. Every row is an exact partition; these are accepted proof-stage executions, not new manuscript experiments.

Traversal and extremum	Seeds	Returns	Difference exits	Height exits
Forward, positive	74,866	25,907	30,335	18,624
Inverse, negative	74,866	25,907	30,335	18,624
Inverse, positive	74,866	25,907	30,344	18,615
Inverse, both signs	149,732	51,814	60,679	37,239

The two unmatched words are

$$\begin{aligned} a = -13 : & \quad (-4, -2, -3, -3, -2), \\ a = -2 : & \quad (-2, -1, 0, 0, -1, -2, 0, -1, 0). \end{aligned}$$

Their first-return lengths are five and nine. The independently reconstructed output agrees with every one of the primary 25 851 records, not only its aggregate counts. The nine-seed exchange between inverse exit types in Table 3 is not a mismatch: opposite directions can first cross different rejecting boundaries. The longest observed walks were 12 forward and 18 backward; neither number was an input. Likewise the observed seed forcing range $[-503, 497]$ was an output, not a prescribed parameter interval.

Corollary 4.2, Lemma 5.1, and this complete exact residual outcome prove exhaustion of the table. The remaining sections verify its existence, least-period, disjointness and level-count assertions without relying on a finite sample of its unbounded parameters.

6 Existence, least periods and oriented representatives

For each row of Table 2, substitute its displayed entries and parameter into $x_{i+3} + x_i - x_{i+1}x_{i+2} - a$. The result is identically zero at every wraparound phase. This is a polynomial identity in the displayed free integer variables, so it establishes existence on the full parameter range, not just on the finite core. Substitution of the initial triple in K_a gives exactly the displayed level. Invariance then gives that same value at all other phases. The preserved separate symbolic check verifies these recurrence and level identities at all 55 phases across the ten rows, in addition to the full invariant identity; the identities themselves are immediate from the printed words.

We give all period and identification arguments, since a periodic word alone is not a disjoint list of native cycles. For F_1 and F_2 , Lemma 3.1 already gives the least period, and $u < v$ chooses the unique ordering of an alternating pair. The length-three word $(-2, -2, t)$ is constant only at $t = -2$. Thus $t \neq -2$ gives exactly period three; its parameter is uniquely recovered from $a = 2t - 4$.

For F_4 , shifting by two positions exchanges t and $a + 1 - t$. A shorter period is possible exactly when $2t = a + 1$, giving an alternating word. At fixed a , the strict inequality $2t < a + 1$ removes that case and chooses one representative of each exchange. Any further cyclic identification must preserve the alternating positions carrying -1 ; if one of the free entries also equals -1 , checking the four positions gives the same exchange and no extra identification.

The F_5 word has prime length five and is never constant, as it contains both 0 and -1 . For $t \notin \{0, -1\}$ its displayed occurrence of -1 is unique. A cyclic identification of two such words must align that occurrence, forcing equal parameters. The words at $t = 0$ and $t = -1$ are rotations of one another and are the sole duplication. Omitting $t = -1$ therefore gives the stated range. In particular, parameters t and $-1 - t$ generally give different oriented cycles, exchanged by time reversal. The exceptional five-cycle E_5 is nonconstant and lies at the different parameter $a = -13$.

For F_6 , the condition $t \neq 0$ excludes every proper divisor 1, 2, 3 of six. A shift by three positions exchanges t with $-t$, and the two nonzero positions show that these are the only possible parameter identifications. The choice $t \geq 1$ is consequently exact.

For F_8 , a shift by four positions exchanges t with $-t - 2$. Period four would force $t = -1$, and periods one or two would imply period four. The excluded value $t = -1$ is already F_4 at level 1. For $t \geq 0$, the positions carrying -1 occur precisely four apart, so a cyclic identification can only give the stated exchange. Exactly one member of each nondegenerate pair has $t \geq 0$.

For F_{12} with $m \geq 2$, compare entries with zero-based indices. Periods one or two are impossible because the entries at positions 0 and 4 are 1 and -1 . Period three is excluded by positions 1 and

4; period four by positions 0 and 4; period six by positions 1 and 7, which are m and $1 - m$. For $m = 1$ the word is

$$(1, 1, 1, 0, -1, -1, 1, 0, 1, -1, -1, 0).$$

Positions 0, 3 exclude period three, positions 0, 4 exclude period four, and positions 1, 7 exclude period six. Periods one and two are again excluded by positions 0, 4. These are every proper divisor of twelve. Since $m^2 - m + 2$ is strictly increasing for integers $m \geq 1$, different standard parameters cannot describe the same cycle. This is a direct least-period argument; no unprinted appeal to the unforced predecessor is required.

The length-nine exceptional word has different entries in positions 0 and 3, so it has neither period three nor period one. It therefore has least period nine. The reversed E_5 word is its rotation by one position, and the reversed E_9 word is its rotation by six. Thus no additional reverse-oriented exceptions are hidden by the list.

Finally, rows of different least periods cannot overlap, and the two rows of length five have different forcing parameters. Together with Section 5, these observations complete every assertion of Theorem 2.1.

7 Exact level counts and the native dynamical zeta function

Fix integers a, k . Let $c_\ell(a, k)$ be the number of oriented cycles of least period ℓ on the ordinary integer level set

$$X_{a,k}(\mathbb{Z}) = \{(x, y, z) \in \mathbb{Z}^3 : K_a(x, y, z) = k\}.$$

All lengths outside (5) have count zero. The following explicit tests remove every remaining orbit search. Here a positive square means h^2 for an integer $h > 0$; zero is excluded whenever positivity is specified.

Proposition 7.1 (Finite arithmetic tests). *The counts $c_\ell(a, k)$ are given by the following rules.*

1. c_1 is the number of distinct integers $r = 1 \pm \sqrt{1 - a}$ satisfying $k = r^2(2r - 3)$. If $1 - a$ is not a nonnegative integer square, the count is zero; if the square root is zero, count the common root once.
2. Let P run over the distinct integer roots of

$$P^2 + (a - 3)P - k = 0. \tag{21}$$

Put $S = P + a$. Each root contributes one to c_2 exactly when $S^2 - 4P = h^2 > 0$ with $h \equiv S \pmod{2}$. Otherwise it contributes zero.

3. $c_3 = 1$ exactly when a is even, $t = (a + 4)/2 \neq -2$, and $k = 8 - (t - 4)^2$; otherwise $c_3 = 0$.
4. $c_4 = 1$ exactly when

$$(a + 1)(a - 7) + 4k = h^2 > 0, \quad h \equiv a + 1 \pmod{2}; \tag{22}$$

otherwise $c_4 = 0$.

5. At $a = -1$, the F_5 contribution is one if $k = 0$, two if $k > 0$ and $4k + 1$ is an integer square, and zero otherwise. Add one precisely for $(a, k) = (-13, -64)$. There is no other contribution to c_5 .

6. $c_6 = 1$ exactly when $a = 0$ and k is a positive integer square.
7. $c_8 = 1$ exactly when $a = -1$ and $k - 1$ is a positive integer square.
8. $c_9 = 1$ exactly when $(a, k) = (-2, -1)$.
9. $c_{12} = 1$ exactly when $a = 0$ and $4k - 7$ is a positive integer square.

In the last four rules the count is zero when the stated condition fails.

Proof. The fixed-point relation is $(r - 1)^2 = 1 - a$, giving the first test. For an alternating pair put $S = u + v$ and $P = uv$. The parameter and level equations become

$$a = S - P, \quad k = P(S - 3) = P(P + a - 3).$$

This proves (21). A distinct integral unordered pair exists exactly when its discriminant is a positive square with the required parity; then

$$u = (S - h)/2 < v = (S + h)/2.$$

Distinct values of P give distinct pairs, since $S = P + a$. This formulation remains valid at $a = 1$ and does not divide by $u - 1$ or lose a factor-zero case.

The F_3 relation uniquely fixes t when a is even. For F_4 , solving $k = t^2 - (a + 1)t + 2a + 2$ gives discriminant $(a + 1)(a - 7) + 4k$. Its smaller root is $(a + 1 - h)/2$; it is integral and satisfies $2t < a + 1$ exactly under (22). Thus there is one cycle, not two, when that test succeeds.

For F_5 , $k = t(t + 1) \geq 0$ for integral t . At $k = 0$ the two roots $0, -1$ give the same cyclic word and the specified range retains only 0. At $k > 0$ the square test $4k + 1 = (2t + 1)^2$ gives two distinct integer parameters, neither excluded; they give the two distinct reversed oriented cycles and must both be counted. The additional exceptional contribution is read at its separate parameter and level.

The formulas $k = t^2$ with $t \geq 1$ and $k - 1 = (t + 1)^2$ with $t \geq 0$ give the sixth and seventh tests. The eighth is the unique E_9 row. Finally $k = m^2 - m + 2$ is equivalent to $4k - 7 = (2m - 1)^2$. A positive square congruent to 1 modulo 4 has odd positive root h , yielding the unique $m = (1 + h)/2 \geq 1$. The disjointness already proved means that no further identifications enter these tests. $\square$

Corollary 7.2 (Ordinary fixed points and zeta). *For all integers a, k and $n \geq 1$,*

$$\# \text{Fix}(T_a^n | X_{a,k}(\mathbb{Z})) = \sum_{\substack{\ell \in \mathcal{L} \\ \ell | n}} \ell c_\ell(a, k). \quad (23)$$

The Artin–Mazur dynamical zeta series in the ordinary time variable τ is the finite rational product

$$\zeta_{a,k}(\tau) := \exp \left(\sum_{n \geq 1} \frac{\# \text{Fix}(T_a^n | X_{a,k}(\mathbb{Z}))}{n} \tau^n \right) = \prod_{\ell \in \mathcal{L}} (1 - \tau^\ell)^{-c_\ell(a,k)}. \quad (24)$$

Proof. Each arithmetic test admits at most finitely many parameters, so $X_{a,k}(\mathbb{Z})$ has finitely many periodic points. An oriented cycle of length ℓ contributes its ℓ distinct points to $\text{Fix}(T_a^n)$ exactly when $\ell \mid n$, proving (23). For each such cycle its contribution to the logarithm of the zeta series is $\sum_{j \geq 1} \tau^{j\ell}/j = -\log(1 - \tau^\ell)$. Summing over the finitely many cycles proves (24) as a formal power-series identity. $\square$

There is no analogous unrestricted all-level finite-count assertion here. For every fixed a , the infinitely many integers t with $2t < a + 1$ give distinct four-cycles. In particular the all-level fixed set of T_a^4 is infinite. Fixing the invariant level is indispensable for the finite counts in (23).

8 Scope, evidence and limitations

The classification concerns ordinary integer triples for the specified polynomial automorphism, including singular ordinary points on its cubic levels. It makes no claim about all rational points, whole-group finite orbits, complex scheme-theoretic fixed loci, multiplicity weights, or an altered compactified or induced return clock. The residual computation is conditional only on the printed analytic reduction and exact integer execution; it is not a proof based on checking a guessed height or period limit. Its finite outputs do not enumerate the unbounded families globally.

The polynomial identities and all signed analytic branches are included in the article. The finite part was checked by a separately written inverse-time implementation, which matched every complete directed-state set. This is computational corroboration with a disclosed implementation boundary, not a formal verification in a proof assistant. Hashes identify the accepted files but do not by themselves establish their mathematical correctness. The assertion caveat for the primary Python implementation is stated in Appendix A. The unchanged source files, complete cycle outputs and full execution summaries accompany the manuscript so the finite claim is reproducible.

The factors in (24) are dynamical cycle factors. They are not Euler factors of an elliptic curve, an arithmetic L -function, or evidence for a root-number assertion. No transfer from source dynamical arithmetic to such a target is asserted.

Preparation and review disclosure. The manuscript and proof artifacts were prepared with AI assistance. The analytic and computational checks cited here are internal current-team reviews, not external human peer review. New manuscript-transcription reviews and release checks are tracked separately from the accepted mathematical executions. Anonymous authorship is maintained; no institution, journal acceptance or external reviewer identity is implied.

References

- [1] Anonymous. Integral periodic points of the Fibonacci trace map, 2026. Research manuscript C413, 6 September 2026. No journal publication asserted.
- [2] Serge Cantat and Frank Loray. Dynamics on character varieties and Malgrange irreducibility of Painlevé VI equation. *Annales de l’Institut Fourier*, 59(7):2927–2978, 2009. doi: 10.5802/aif.2512. URL https://aif.centre-mersenne.org/item/AIF_2009__59_7_2927_0/.
- [3] Andrew N. W. Hone. Diophantine non-integrability of a third order recurrence with the Laurent property. *Journal of Physics A: Mathematical and General*, 39(12):L171–L177, 2006. doi: 10.1088/0305-4470/39/12/L01.
- [4] Stephen Humphries. Points with finite orbits for trace maps. arXiv:1611.02743v1, 2016. URL <https://arxiv.org/abs/1611.02743v1>. Preprint.
- [5] John A. G. Roberts. Escaping orbits in trace maps. *Physica A: Statistical Mechanics and its Applications*, 228(1–4):295–325, 1996. doi: 10.1016/0378-4371(95)00428-9.
- [6] John A. G. Roberts and Michael Baake. Trace maps as 3D reversible dynamical systems with an invariant. *Journal of Statistical Physics*, 74(3–4):829–888, 1994. doi: 10.1007/BF02188581.

A Exact implementation and preserved evidence

A.1 Computational contract and provenance

The primary script below is the unchanged implementation used for the accepted positive-extremum run. It uses Python integers, not floating-point approximations, and its only maximum is the proved amplitude constant 100. The infinite-looking orbit loop terminates by Lemma 5.1. The script contains assertions: reproducing its historical validation requires ordinary Python execution, *without* the optimization flag -O. The independent implementation uses explicit exceptions and a separately implemented seed and state-set algorithm as specified completely in Section 5.

The accompanying `supplement/` tree preserves the relative layout needed by the independent script: its primary input directory is `integral_return/`, and its own directory is `integral_return_review/finite_core/`. The source proof, both certifiers, both complete cycle JSONL files, both execution summaries, the separate symbolic checker, and the internal proof-review reports are retained unchanged. Running the scripts writes their outputs beside their sources; to reproduce without replacing the accepted receipts, first copy the supplement to a fresh scratch directory and run there. The raw 25 851 records are machine-readable witnesses; their seed generation, stopping criterion, matching and semantic comparison have all been specified in the article.

The independent run used Python 3.12.3. Its recorded runtime was 3.37218 seconds; the primary recorded runtime was 0.834775 seconds. These are historical provenance, not benchmark claims or complexity bounds. The separate symbolic receipt used Python 3.12.3 and SymPy 1.14.0 and checked the ten rows at all 55 wraparound phases for each recurrence and level identity, the invariant identity and four algebraic count-reduction identities. None of these accepted mathematical programs was rerun during manuscript preparation.

The full primary and independent summaries retain the exact seed dispositions, per-amplitude independent counts and all observed bounds. In the primary JSONL, each record contains `a`, `least_period`, `word`, and `family`; the word is the least cyclic rotation and the records are sorted by parameter and word. The independent output instead reconstructs native words from least triples. Different byte encodings therefore need not share a hash; agreement was checked on the full oriented mathematical data.

A.2 Accepted byte identifiers

The following SHA-256 values identify the historical evidence. They are not substitutes for the reduction proof or the complete finite comparison.

Primary certifier

750b4dbb54cacd1df11cc3929ed436cf8de9877048545f212cecc9bc03b75330

Primary cycle JSONL

352ffd5b5b188e32c347a4083559680e5d1bad12c8e6c6d7cd3dad6324935805

Primary summary

7000a021df6d38d82e14f6a22b3b764026e9545881aa4b7247cd041e5cf4af0b

Independent certifier

8bd343b599f9ed6c605b255dfe60b544a81375975083fd5d7ffb2b18a1d6c7c3

Independent cycle JSONL

0432cd5d8c35fe3cf9fdc0027451bf2e85d04eb880da6da1c03457571fffe35b

Separate symbolic checker

be9ca47f3b7c8f94b43b25d6857f823f35f004407eff71807770dadf6dac40bd

A.3 Complete primary certifier

The following listing includes the full source, including family subtraction, primitive-word checks, both orientation restoration and summary generation. The inverse-time script is supplied in machine-readable form; its exact mathematical specification is in Section 5.

```
1  #!/usr/bin/env python3
2  """Exhaust the analytically proved IR1 core, using exact Python integers.
3
4  No chosen parameter bound, period bound, numerical tolerance or random
5  sample occurs. D_MAX=100 is the theorem's proved reduction constant.
6  The input contract and termination argument are in IR1_PROOF.md, S4.
7  Generated JSON files are evidence, not independent proof of the reduction.
8  """
9  from collections import Counter
10 from hashlib import sha256
11 import json
12 from pathlib import Path
13 from time import monotonic
14
15 D_MAX = 100
16 OUT = Path(__file__).resolve().parent
17
18
19 def canonical(word):
20     return min(word[i:] + word[:i] for i in range(len(word)))
21
22
23 def family(a, word):
24     n = len(word)
25     if n <= 2:
26         return "F2"
27     for j in range(n):
28         w = word[j:] + word[:j]
29         t = w[1]
30         if n == 3 and w[:2] == (-2, -2) and a == 2*w[2]-4:
31             return "F3"
32         if n == 4 and w[0] == w[2] == -1 and w[3] == a+1-t:
33             return "F4"
34         if n == 5 and a == -1 and w == (-1, t, 0, 0, -1-t):
35             return "F5"
36         if n == 6 and a == 0 and w == (0, 0, w[2], 0, 0, -w[2]):
37             return "F6"
38         if n == 8 and a == -1 and w == (-1,t,1,t,-1,-t-2,1,-t-2):
39             return "F8"
40         if n == 12 and a == 0 and t >= 1:
41             if w == (1,t,1,t-1,-1,-t,1,1-t,1,-t,-1,t-1):
42                 return "F12"
43     return "EXCEPTION"
44
45
46 def verify_cycle(a, word):
47     n = len(word)
48     assert n > 0
49     for i in range(n):
50         assert word[(i+3) % n] == word[(i+1) % n]*word[(i+2) % n]+a-word[i]
51     assert all(any(word[i] != word[(i+k) % n] for i in range(n)) for k in range(1,n))
52
53
54 def run():
55     start = monotonic()
56     counts = Counter()
57     exits = Counter()
```



```

58 cycles = set()
59 max_steps = 0
60 # No change of a or D during an orbit. Every loop is explicitly finite
61 # except its orbit iteration, whose termination follows by injectivity.
62 for D in range(1, D_MAX+1):
63     lo, hi = -3*D-1, 3*D-1
64     for u in range(lo, hi-D+1):
65         v = u+D
66         for s in (-3,-2,-1,0,1):
67             if not lo <= s <= hi:
68                 continue
69             center_sum = (s+1)*D
70             p_lo = max(-D, center_sum-D)
71             p_hi = min(D, center_sum+D)
72             if u+1:
73                 bp = (2*D)//abs(u+1)
74                 p_lo, p_hi = max(p_lo,-bp), min(p_hi,bp)
75             if v+1:
76                 bq = (2*D)//abs(v+1)
77                 p_lo, p_hi = max(p_lo,center_sum-bq), min(p_hi,center_sum+bq)
78             for p in range(p_lo,p_hi+1):
79                 q = center_sum-p
80                 assert abs(p)<=D and abs(q)<=D
81                 assert abs((u+1)*p)<=2*D and abs((v+1)*q)<=2*D
82                 a = v-s*u+s-p
83                 seed = (u,s,v)
84                 x,y,z = seed
85                 word = []
86                 counts["seeds"] += 1
87                 while True:
88                     if not (lo<=x<=hi and lo<=y<=hi and lo<=z<=hi):
89                         exits["height"] += 1
90                         break
91                     if abs(z-x)>D:
92                         exits["difference"] += 1
93                         break
94                     word.append(x)
95                     x,y,z = y,z,y*z+a-x
96                     if (x,y,z) == seed:
97                         w = tuple(word)
98                         verify_cycle(a,w)
99                         # The extremal normalization may reverse time.
100                         # Restore both native orientations explicitly.
101                         for candidate in (w, tuple(reversed(w))):
102                             verify_cycle(a,candidate)
103                             cycles.add((a,canonical(candidate)))
104                         counts["returning_seeds"] += 1
105                         break
106                 max_steps = max(max_steps,len(word))
107 ordered = sorted(cycles)
108 rows = []
109 family_counts = Counter()
110 exceptions = []
111 for a,w in ordered:
112     kind = family(a,w)
113     row = {"a":a,"least_period":len(w),"word":list(w),"family":kind}
114     rows.append(row)
115     family_counts[kind] += 1
116     if kind == "EXCEPTION":
117         exceptions.append(row)
118 assert counts["seeds"] == counts["returning_seeds"] + sum(exits.values())
119 raw = "".join(json.dumps(row,separators=(",", ":"),sort_keys=True)+"\n" for row in rows)
120 (OUT/"IR1_CORE_CYCLES.jsonl").write_text(raw,encoding="utf-8")
121 summary = {
122     "D_MAX":D_MAX,
123     "seed_rule":"IR1_PROOF.md section 4; neighbor bounds included",
124     "parameter_cutoff":None,"period_cutoff":None,"exact_integer_arithmetic":True,
125     "all_counts":dict(counts),"exit_counts":dict(exits),

```

```

126         "max_orbit_steps_before_return_or_certified_exit":max_steps,
127         "distinct_oriented_cycles_in_completed_seed_output":len(rows),
128         "family_cycle_counts_in_finite_certificate_only":dict(sorted(family_counts.items())),
129         "exceptions":exceptions,
130         "cycle_jsonl_sha256":sha256(raw.encode()).hexdigest(),
131         "code_sha256":sha256(Path(__file__).read_bytes()).hexdigest(),
132         "elapsed_seconds":round(monotonic()-start,6),
133     }
134     (OUT/"IR1_CORE_SUMMARY.json").write_text(json.dumps(summary,indent=2,sort_keys=True)+"\n",encoding="utf-8"
135     )
136     print(json.dumps(summary,indent=2,sort_keys=True))
137
138 if __name__ == "__main__":
139     run()

```